

Network Management and Adminsitration (IE2110)
2nd Year, 2nd Semester

Assignment 01

Zimbra

Submitted to

Sri Lanka Institute of Information Technology

<IT24103400>

<Jayantha . K. P>

In partial fulfillment of the requirements for the

Bachelor of Science Special Honors Degree in Information Technology

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Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

Introduction

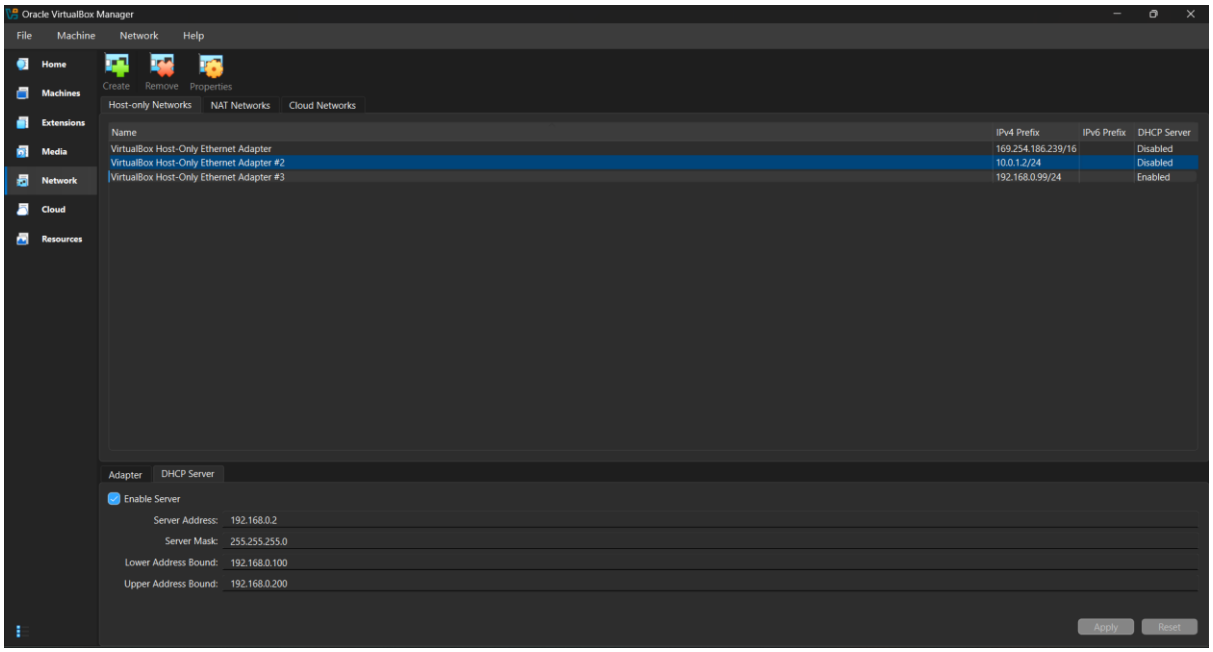
This assignment required the deployment and validation of a Linux-based network environment capable of delivering several core administrative services within a controlled virtual laboratory. The completed environment covers the full set of required functions: initial virtual machine preparation, network configuration, dynamic IP address assignment, electronic mail services, centralized directory and authentication services, and web publishing. The implementation used the assignment-specific values of network 192.168.0.0/24, DHCP range 192.168.0.100 - 192.168.0.200, mail domain mail.it05.local, and Active Directory domain it05.local.

The assignment brief contains platform consistency. It summery requires Zimbra and samba installed and configured and creates the mails on Ubuntu Servers and the web server in finished on CentOS 10 version

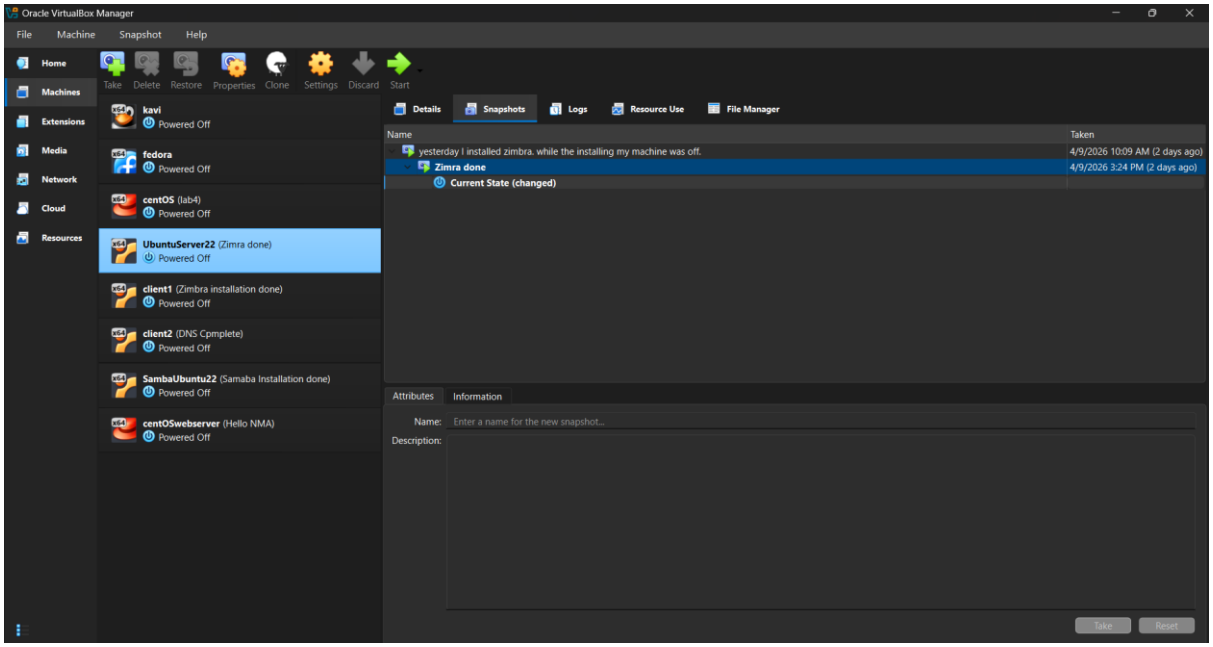
After the configuration phase, the environment was tested to verify that all major services were functioning correctly. The validation confirmed successful DHCP lease allocation, correct mail account creation and email communication through Zimbra, proper domain and user management through Samba Active Directory, successful domain access from the Ubuntu client machine, and correct client-side access to the hosted web page. Therefore, the completed system demonstrates a fully operational Linux-based network environment capable of delivering the required core administrative services in accordance with the assignment objectives.

Part A – Initial Setup

Virtual network preparation

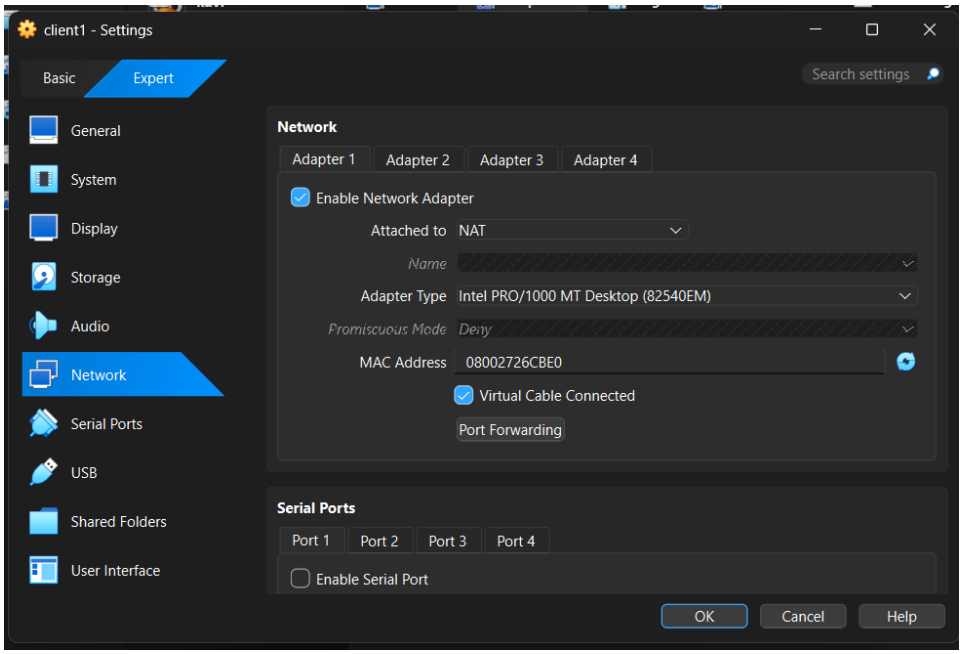


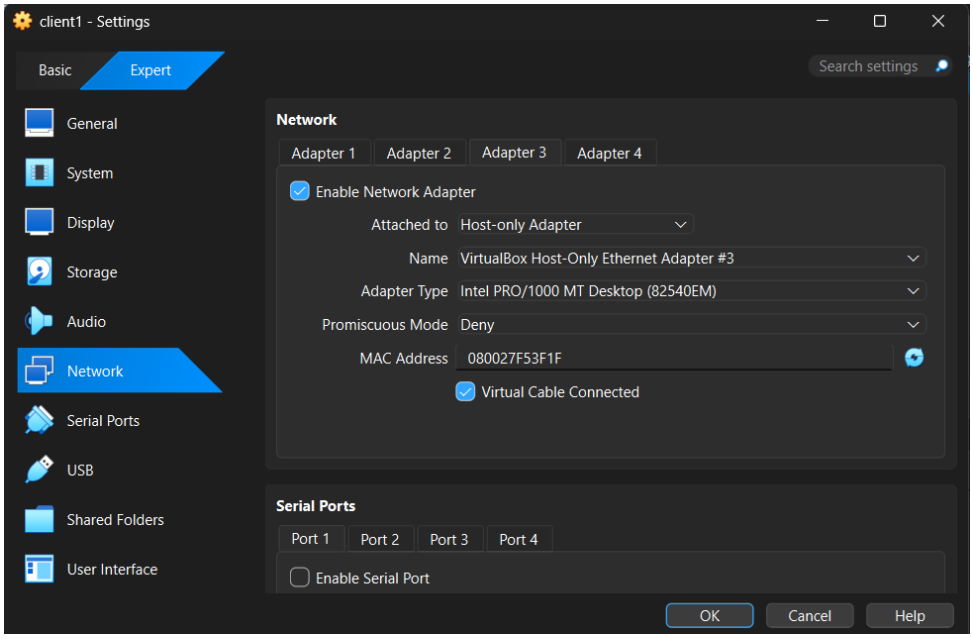
The following is the diagram of the VirtualBox host-only network setup that has been created for the internal lab network. Within this set up, two Ubuntu servers have been setup for Zimbra mail services and Samba Active Directory Services respectively, while CentOS server is set to operate as web server. Since there was a problem with Adapter 2, an additional Adapter, namely, Adapter 3 has been added which has been set as host-only network interface. Adapter 3 has been configured with IP address 192.168.0.2 which became the base address of the internal lab network.



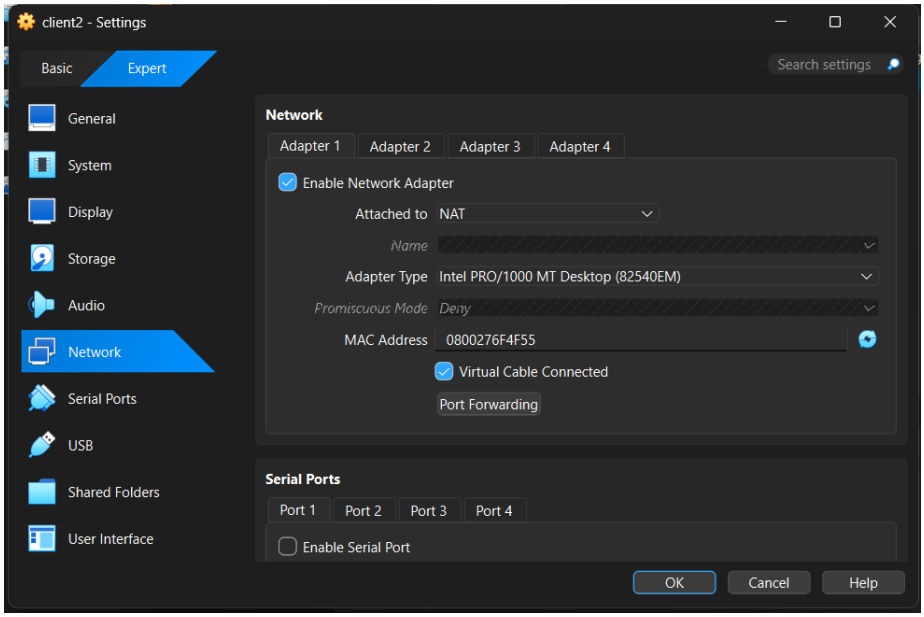
Above figure show the virtual machine inventory created for the assignment implementation It confirms that separate machines were prepared for DHCP, Samba Active Directory, Zimbra, and client validation and separated servers for Zimbra,Samba and Web servers.

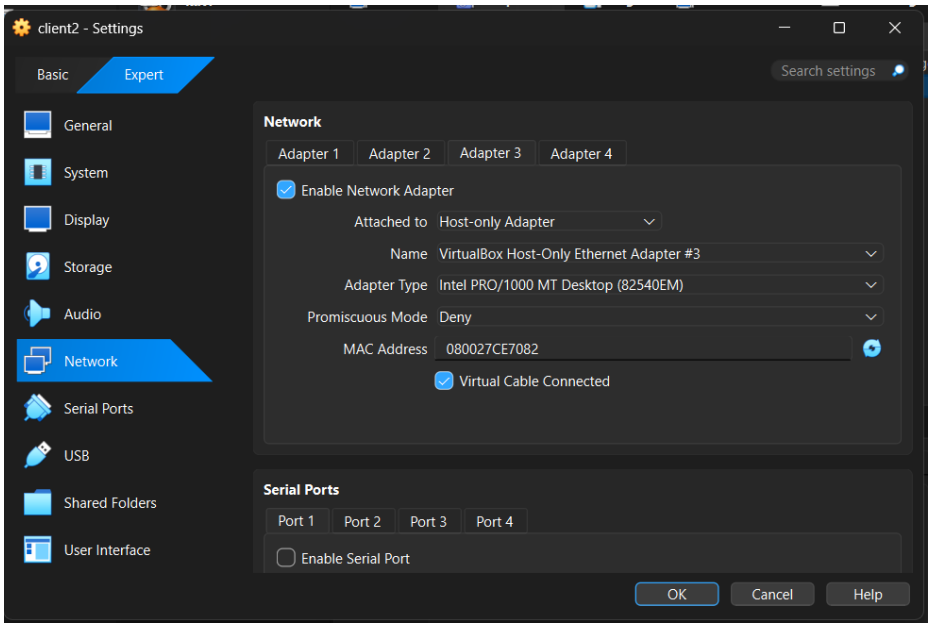
Ubuntu client 1



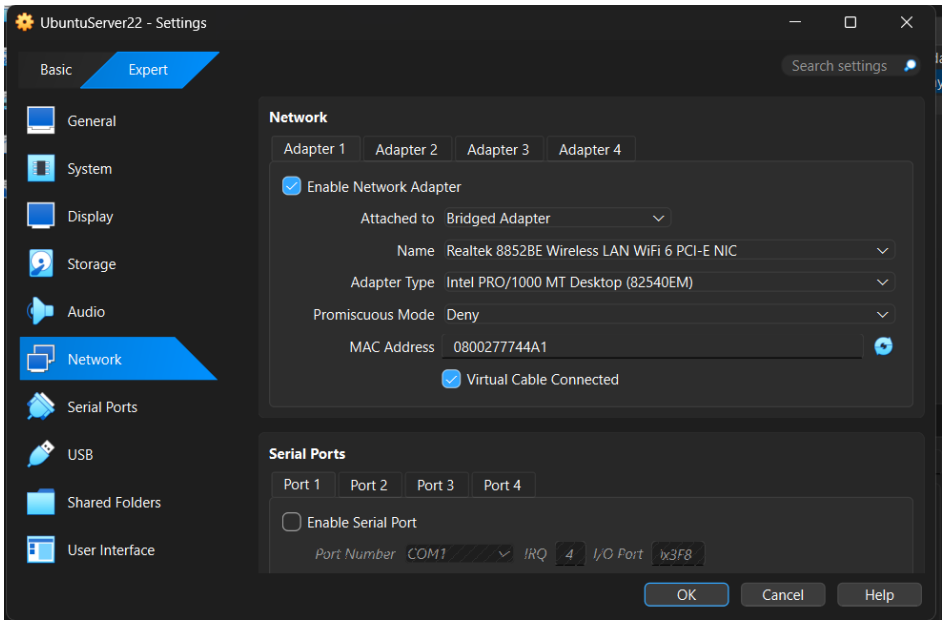


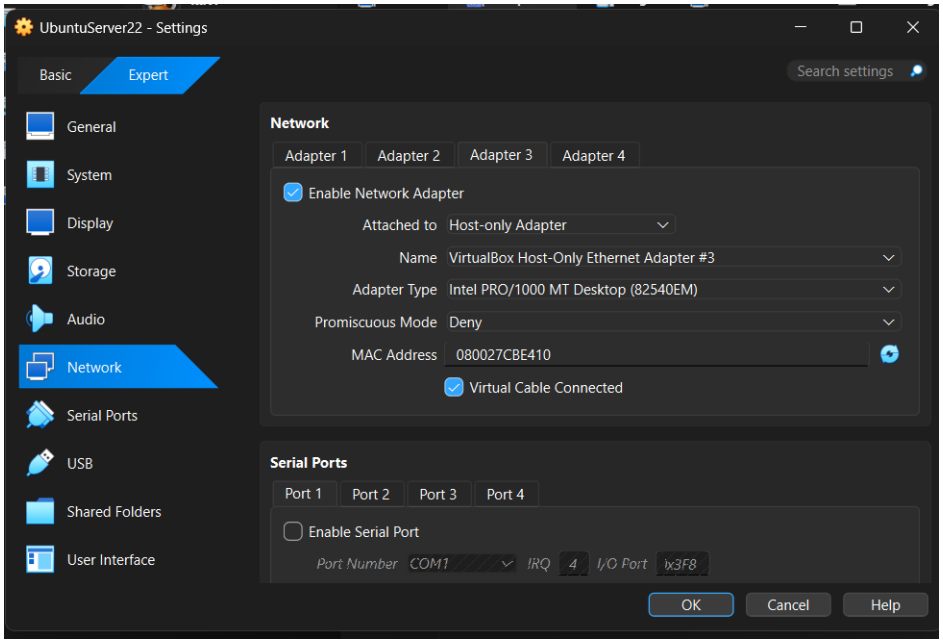
Ubuntu client 2



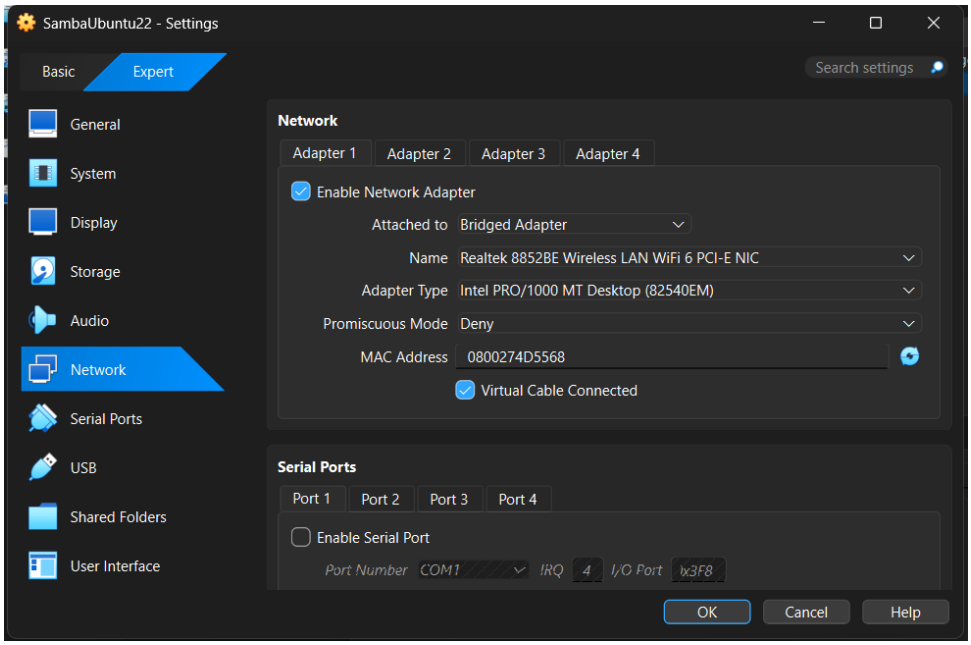


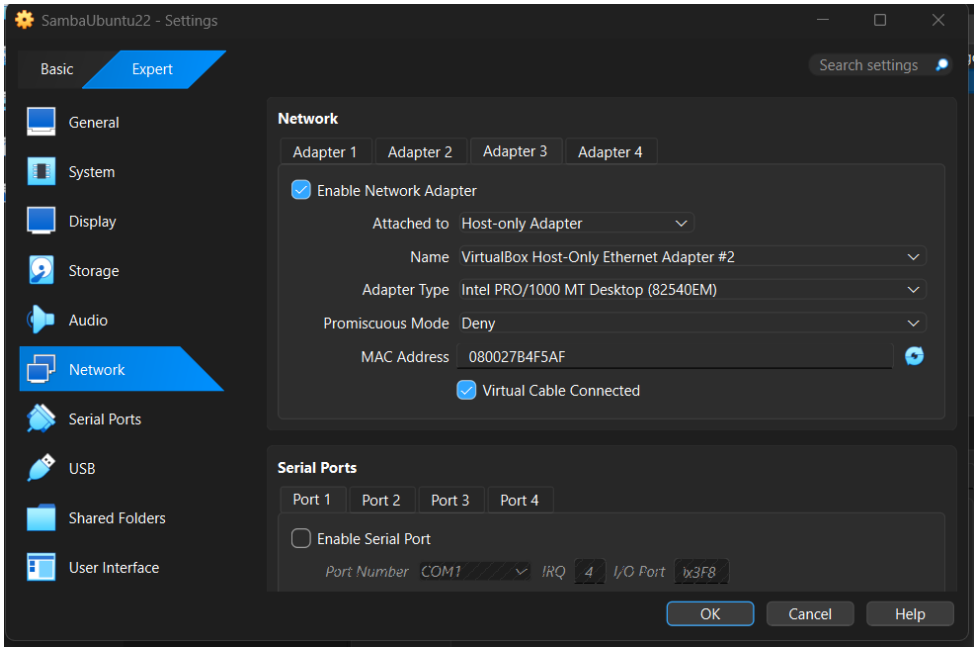
UbuntuZimbra



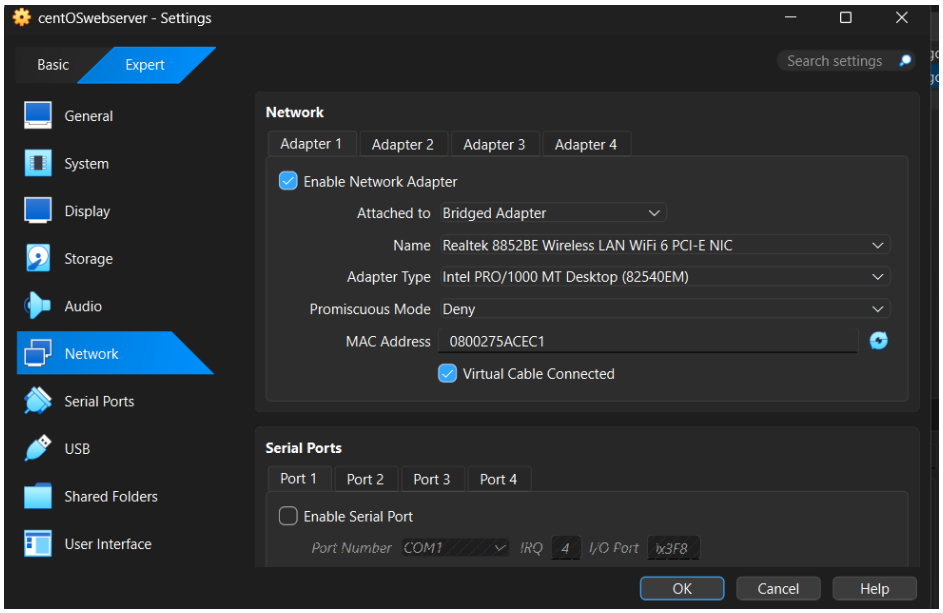


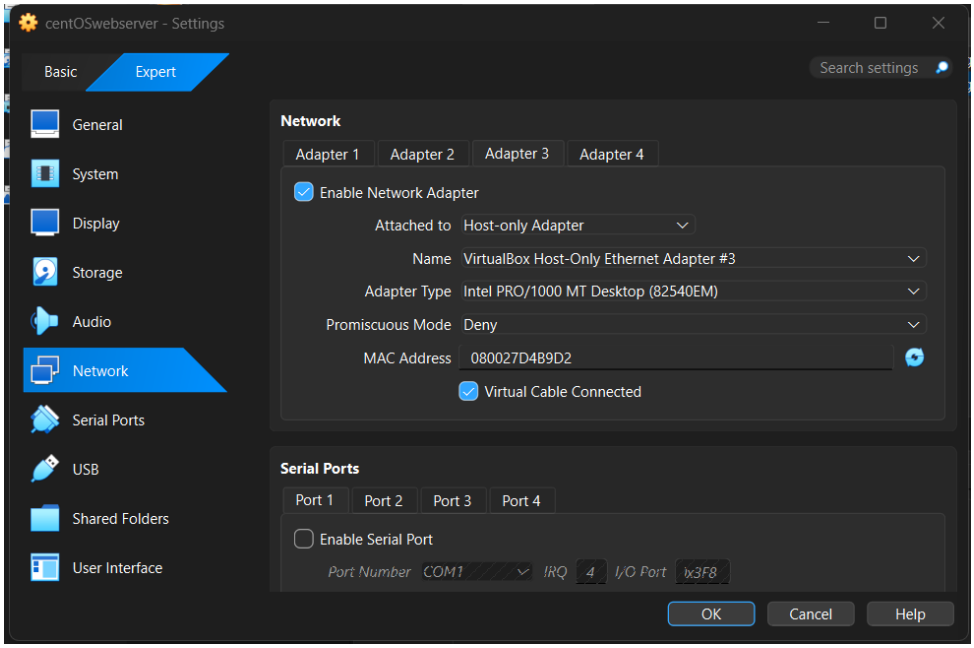
UbuntuSamba





CentOs Web Server





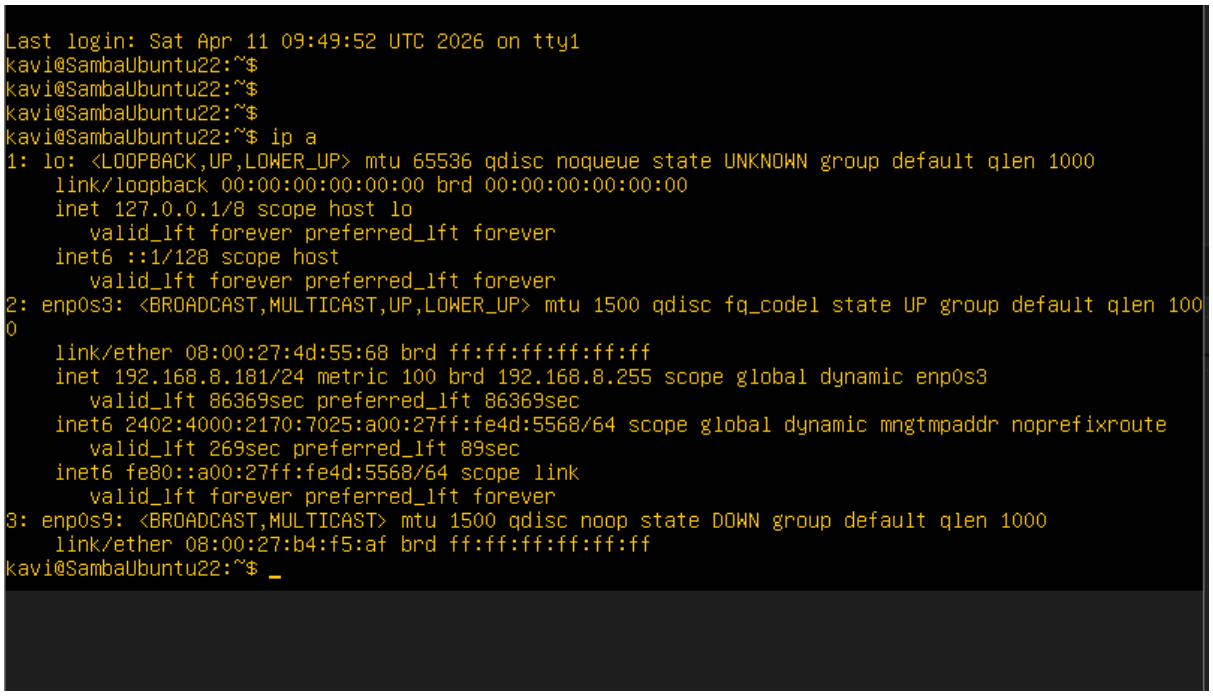
For the two client virtual machines, Adapter 1 was configured in NAT mode to allow internet connectivity for software installation, updates, and general external access. Adapter 3 was configured as Host-Only so that both clients could communicate within the internal laboratory network. For the server machines, Adapter 1 was set to Bridged mode to enable direct network connectivity and support SSH access from the host or other devices on the physical network. In addition, Adapter 3 was configured as Host-Only to connect the servers to the isolated internal lab segment used for service testing and client-server communication.

Interface and connectivity validation

UbuntuZimbra



UbuntuSamba



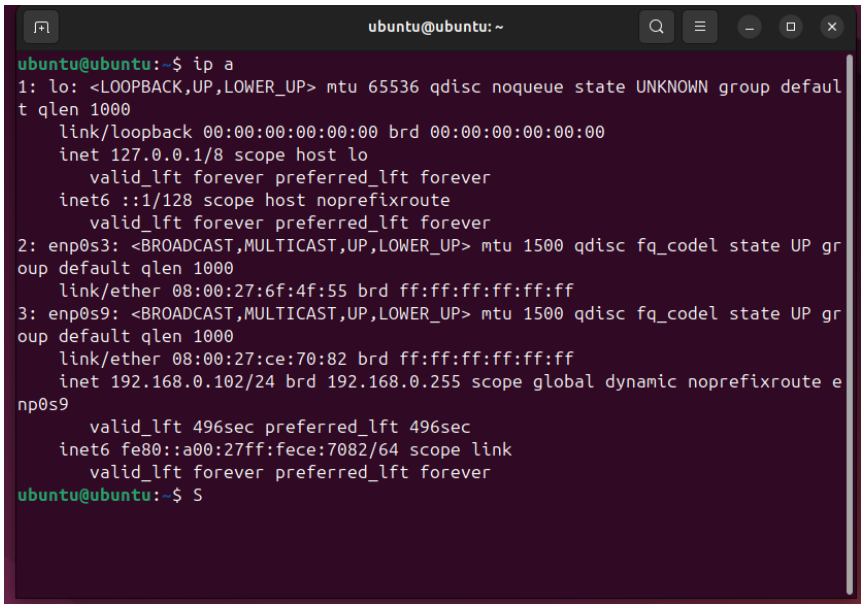
CentOsWebServer

```
[kavi@localhost ~]$
[kavi@localhost ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:ce:c1 brd ff:ff:ff:ff:ff:ff
    altname enx0800275acec1
    inet 192.168.8.164/24 brd 192.168.8.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86083sec preferred_lft 86083sec
    inet6 2402:4000:2170:7025:a00:27ff:fe5a:cec1/64 scope global dynamic noprefixroute
       valid_lft 295sec preferred_lft 115sec
    inet6 fe80::a00:27ff:fe5a:cec1/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d4:b9:d2 brd ff:ff:ff:ff:ff:ff
    altname enx080027d4b9d2
    inet 192.168.0.1/24 brd 192.168.0.255 scope global noprefixroute enp0s9
       valid_lft forever preferred_lft forever
    inet6 fe80::fdba:aa15:8caa:16ff/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[kavi@localhost ~]$
```

Ubuntu Client 1

```
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:36:6f:ca brd ff:ff:ff:ff:ff:ff
3: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:c7:88:2f brd ff:ff:ff:ff:ff:ff
    inet 192.168.61.102/24 brd 192.168.61.255 scope global dynamic noprefixroute
 enp0s9
       valid_lft 328sec preferred_lft 328sec
    inet6 fe80::a00:27ff:fec7:882f/64 scope link
       valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$
```

Ubuntu Client 2

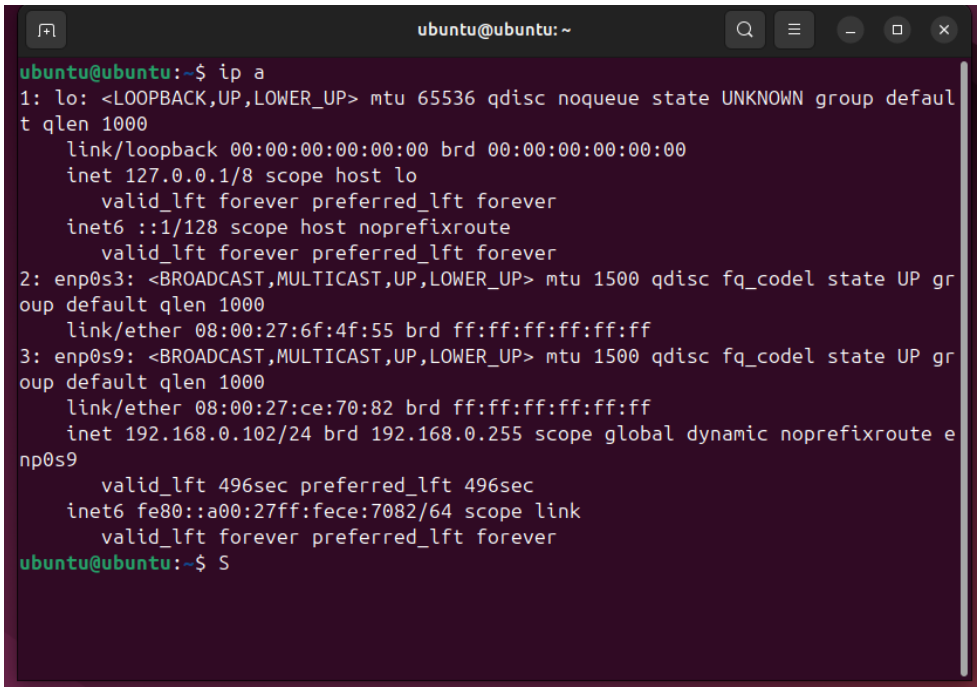


Part B – DHCP Server

```
kavi@UbuntuServer22:~$ sudo systemctl restart isc-dhcp-server
sudo: unable to resolve host UbuntuServer22: Name or service not known
[sudo] password for kavi:
kavi@UbuntuServer22:~$
kavi@UbuntuServer22:~$ sudo systemctl enable isc-dhcp-server
sudo: unable to resolve host UbuntuServer22: Name or service not known
Synchronizing state of isc-dhcp-server.service with SysV service script with
 /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable isc-dhcp-server
kavi@UbuntuServer22:~$ sudo systemctl status isc-dhcp-server
sudo: unable to resolve host UbuntuServer22: Name or service not known
● isc-dhcp-server.service - ISC DHCP IPv4 server
     Loaded: loaded (/lib/systemd/system/isc-dhcp-server.service; enabled; >
     Active: active (running) since Sat 2026-04-11 12:43:44 UTC; 1min 4s ago
       Docs: man:dhcpd(8)
   Main PID: 8367 (dhcpd)
      Tasks: 4 (limit: 9384)
     Memory: 4.5M
        CPU: 26ms
     CGroup: /system.slice/isc-dhcp-server.service
             └─8367 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-serv>

Apr 11 12:43:44 UbuntuServer22 dhcpd[8367]: PID file: /run/dhcp-server/dhcp>
Apr 11 12:43:44 UbuntuServer22 dhcpd[8367]: Wrote 0 leases to leases file.
Apr 11 12:43:44 UbuntuServer22 sh[8367]: Wrote 0 leases to leases file.
Apr 11 12:43:44 UbuntuServer22 dhcpd[8367]: Listening on LPF/enp0s9/08:00:2>
Apr 11 12:43:44 UbuntuServer22 sh[8367]: Listening on LPF/enp0s9/08:00:27:c>
Apr 11 12:43:44 UbuntuServer22 sh[8367]: Sending on   LPF/enp0s9/08:00:27:c>
Apr 11 12:43:44 UbuntuServer22 sh[8367]: Sending on   Socket/fallback/fallb>
Apr 11 12:43:44 UbuntuServer22 dhcpd[8367]: Sending on   LPF/enp0s9/08:00:2>
Apr 11 12:43:44 UbuntuServer22 dhcpd[8367]: Sending on   Socket/fallback/fa>
Apr 11 12:43:44 UbuntuServer22 dhcpd[8367]: Server starting service.
lines 1-21/21 (END)
```

I installed kea DHCP configuration on ubuntu server for the assignment subnet.
Subnet – 192.168.0.0
Range – 192.168.0.100 – 192.168.0.200
This figure records the operational state of the Kea DHCP service after activation. It verifies that the DHCP daemon started successfully on the Ubuntu server and was available to answer lease requests from client systems. This is important because a correct configuration file alone does not satisfy the assignment; the service must also be actively running.



Above figure provides the client-side proof of DHCP functionality. It confirms that the Ubuntu client received an address automatically from the CentOS server and that the addressing process moved from manual configuration to dynamic allocation as required. This is important because it demonstrates the real functional outcome of the DHCP deployment from the perspective of the consuming client machine

Part C – Zimbra Mail Server

DNS installation and active

```
kavi@UbuntuServer22:~$ sudo systemctl status bind9
sudo: unable to resolve host UbuntuServer22: Name or service not known
● named.service - BIND Domain Name Server
     Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor pre>
     Active: active (running) since Sat 2026-04-11 12:40:28 UTC; 10min ago
       Docs: man:named(8)
   Main PID: 816 (named)
      Tasks: 20 (limit: 9384)
     Memory: 34.7M
        CPU: 742ms
     CGroup: /system.slice/named.service
             └─816 /usr/sbin/named -u bind

Apr 11 12:40:28 UbuntuServer22 named[816]: zone 0.in-addr.arpa/IN: loaded s>
Apr 11 12:40:28 UbuntuServer22 named[816]: zone 127.in-addr.arpa/IN: loaded>
Apr 11 12:40:28 UbuntuServer22 named[816]: zone it00.local/IN: loaded seria>
Apr 11 12:40:28 UbuntuServer22 named[816]: zone 255.in-addr.arpa/IN: loaded>
Apr 11 12:40:28 UbuntuServer22 named[816]: all zones loaded
Apr 11 12:40:28 UbuntuServer22 named[816]: running
Apr 11 12:40:28 UbuntuServer22 systemd[1]: Started BIND Domain Name Server.
Apr 11 12:40:28 UbuntuServer22 named[816]: managed-keys-zone: Key 20326 for>
Apr 11 12:40:28 UbuntuServer22 named[816]: managed-keys-zone: Key 38696 for>
Apr 11 12:41:00 UbuntuServer22 named[816]: broken trust chain resolving '0.>
lines 1-21/21 (END)
```

Zimbra installation

```
Main menu

   1) Common Configuration:
   2) zimbra-ldap:                          Enabled
   3) zimbra-logger:                        Enabled
   4) zimbra-mta:                           Enabled
   5) zimbra-dnscache:                      Enabled
   6) zimbra-snmp:                          Enabled
   7) zimbra-store:                         Enabled

        +Create Admin User:                 yes
        +Admin user to create:              admin@mail.it00.local
******* +Admin Password                      UNSET
        +Anti-virus quarantine user:        virus-quarantine.kslj4z5sq@mail.it00.local
        +Enable automated spam training:    yes
        +Spam training user:                spam.ie17wqlf_k@mail.it00.local
        +Non-spam(Ham) training user:       ham.iwixlmfu@mail.it00.local
        +SMTP host:                         mail.it00.local
        +Web server HTTP port:              8080
        +Web server HTTPS port:             8443
        +Web server mode:                   https
        +IMAP server port:                  7143
        +IMAP server SSL port:              7993
        +POP server port:                   7110
        +POP server SSL port:               7995
        +Use spell check server:            yes
        +Spell server URL:                  http://mail.it00.local:7780/aspell.php
        +Enable version update checks:      TRUE
        +Enable version update notifications:  TRUE
        +Version update notification email:    admin@mail.it00.local
        +Version update source email:          admin@mail.it00.local
        +Install mailstore (service webapp):   yes
        +Install UI (zimbra,zimbraAdmin webapps): yes
******* +License filename:                    UNSET

   8) zimbra-spell:                         Enabled
   9) zimbra-convertd:                      Enabled
  10) zimbra-proxy:                         Enabled
  11) zimbra-imapd:                         Enabled
  12) Default Class of Service Configuration:
```

```
   8) zimbra-spell:                         Enabled
   9) zimbra-convertd:                      Enabled
  10) zimbra-proxy:                         Enabled
  11) zimbra-imapd:                         Enabled
  12) Default Class of Service Configuration:
  13) Enable default backup schedule:       yes
   s) Save config to file
   x) Expand menu
   q) Quit
```

The above two figures illustrates the final summary of Zimbra Collaboration Suite setup. It is evident that all necessary components of Zimbra such as LDAP, MTA, mailbox storage, proxy, and logging have been enabled successfully. The mail server name has been set as mail.it00.local, and ports for all essential services have been allocated automatically. The screen clearly indicates that the setup process has finally come to its final review step where the settings can be saved and implemented. Despite the unavailability of any license information, the necessary mail server configuration has been created successfully.

```
kavi@UbuntuServer22:~$ ping -c 3 mail.it00.local
PING mail.it00.local (192.168.0.1) 56(84) bytes of data.
64 bytes from mail.it00.local (192.168.0.1): icmp_seq=1 ttl=64 time=0.103 ms
64 bytes from mail.it00.local (192.168.0.1): icmp_seq=2 ttl=64 time=0.142 ms
64 bytes from mail.it00.local (192.168.0.1): icmp_seq=3 ttl=64 time=0.380 ms

--- mail.it00.local ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2033ms
rtt min/avg/max/mdev = 0.103/0.208/0.380/0.122 ms
```

Above Figure below presents the result of the successful connectivity test performed on the mail server by pinging it. The domain name ‘mail.it00.local’ has been successfully converted into an IP address ‘192.168.0.1’. The three ICMP Echo requests have been successfully received back with zero percent packet loss and extremely low latency.

Zimbra Configuration

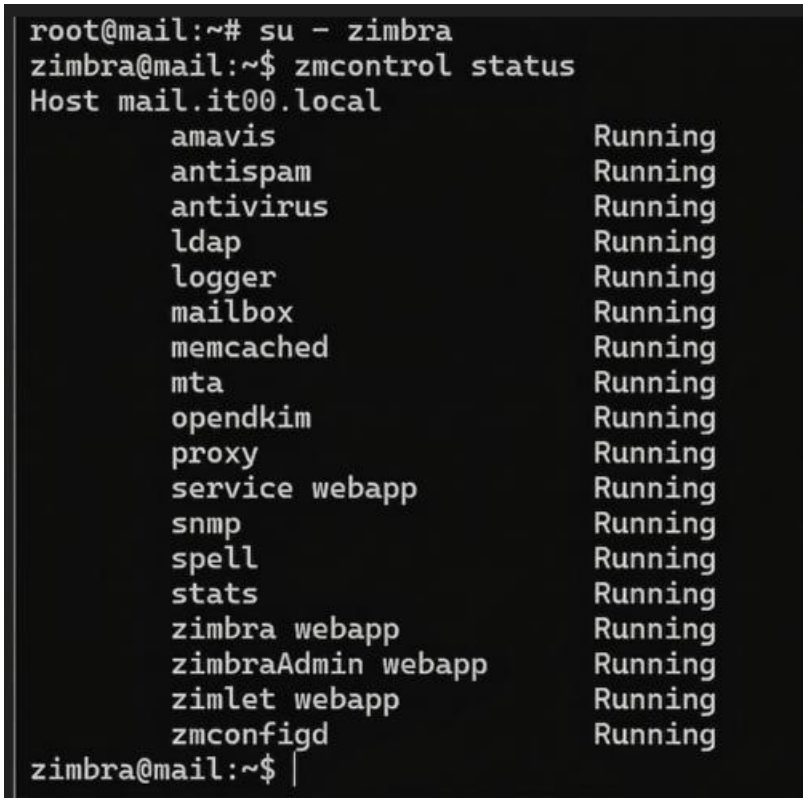
```
Administrator password:
Retype password:
INFO 2026-04-07 21:05:25,297 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #2108: Looking up IPv4 addresses
INFO 2026-04-07 21:05:25,301 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #2125: Looking up IPv6 addresses
INFO 2026-04-07 21:05:25,560 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #2274: Setting up share.ldb
INFO 2026-04-07 21:05:25,666 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #2278: Setting up secrets.ldb
INFO 2026-04-07 21:05:25,738 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #2283: Setting up the registry
INFO 2026-04-07 21:05:26,035 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #2286: Setting up the privileges database
INFO 2026-04-07 21:05:26,185 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #2289: Setting up idmap db
INFO 2026-04-07 21:05:26,286 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #2296: Setting up SAM db
INFO 2026-04-07 21:05:26,310 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #880: Setting up sam.ldb partitions and settings
INFO 2026-04-07 21:05:26,311 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #892: Setting up sam.ldb rootDSE
INFO 2026-04-07 21:05:26,330 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1305: Pre-loading the Samba 4 and AD schema
Unable to determine the DomainSID, can not enforce uniqueness constraint on
local domainSIDs

INFO 2026-04-07 21:05:26,392 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1383: Adding DomainDN: DC=it00,DC=local
INFO 2026-04-07 21:05:26,417 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1415: Adding configuration container
INFO 2026-04-07 21:05:26,448 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1430: Setting up sam.ldb schema
INFO 2026-04-07 21:05:28,762 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1448: Setting up sam.ldb configuration data
INFO 2026-04-07 21:05:28,864 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1489: Setting up display specifiers
INFO 2026-04-07 21:05:30,264 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1497: Modifying display specifiers and extended rights
INFO 2026-04-07 21:05:30,291 pid:5315 /usr/lib/python3/dist-packages/samba/p
```

```
INFO 2026-04-07 21:05:26,392 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1383: Adding DomainDN: DC=it00,DC=local
INFO 2026-04-07 21:05:26,417 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1415: Adding configuration container
INFO 2026-04-07 21:05:26,448 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1430: Setting up sam.ldb schema
INFO 2026-04-07 21:05:28,762 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1448: Setting up sam.ldb configuration data
INFO 2026-04-07 21:05:28,864 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1489: Setting up display specifiers
INFO 2026-04-07 21:05:30,264 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1497: Modifying display specifiers and extended rights
INFO 2026-04-07 21:05:30,291 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1504: Adding users container
INFO 2026-04-07 21:05:30,292 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1510: Modifying users container
INFO 2026-04-07 21:05:30,295 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1513: Adding computers container
INFO 2026-04-07 21:05:30,297 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1519: Modifying computers container
INFO 2026-04-07 21:05:30,297 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1523: Setting up sam.ldb data
INFO 2026-04-07 21:05:30,380 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1553: Setting up well known security principals
INFO 2026-04-07 21:05:30,412 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1567: Setting up sam.ldb users and groups
INFO 2026-04-07 21:05:30,505 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/__init__.py #1575: Setting up self join
Repacking database from v1 to v2 format (first record CN=Port-Name,CN=Schema
,CN=Configuration,DC=it00,DC=local)
Repack: re-packed 10000 records so far
Repacking database from v1 to v2 format (first record CN=siteLink-Display,CN
=401,CN=DisplaySpecifiers,CN=Configuration,DC=it00,DC=local)
Repacking database from v1 to v2 format (first record CN=Allowed RODC Passwo
rd Replication Group,CN=Users,DC=it00,DC=local)
```

```
INFO 2026-04-07 21:05:31,699 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/sambadns.py #1200: Adding DNS accounts
INFO 2026-04-07 21:05:31,767 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/sambadns.py #1234: Creating CN=MicrosoftDNS,CN=System,DC=it00,DC=lo
cal
INFO 2026-04-07 21:05:31,777 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/sambadns.py #1247: Creating DomainDnsZones and ForestDnsZones parti
tions
INFO 2026-04-07 21:05:31,840 pid:5315 /usr/lib/python3/dist-packages/samba/p
rovision/sambadns.py #1252: Populating DomainDnsZones and ForestDnsZones par
titions
Repacking database from v1 to v2 format (first record CN=Deleted Objects,DC=
DomainDnsZones,DC=it00,DC=local)
Repacking database from v1 to v2 format (first record DC=ForestDnsZones,DC=i
t00,DC=local)
```

The above confirms that the Zimbra installation and configuration process was completed successfully.



The above-mentioned image represents the outcome of the zmcontrol status command executed to determine whether the Zimbra mail server was functioning properly after installation. From the output, it can be concluded that the server mail.it00.local was functioning well, and that the majority of Zimbra's important services, namely amavis, antispam, convertd, ldap, logger, mailbox, memcached, mta, proxy, service webapp, zimbra webapp, zimbraAdmin webapp, and zmconfigd were Running. Though antivirus and dnscache services were not running, the essential services required by Zimbra were functioning. It implies that the Zimbra application was up and running and was prepared to be tested in the next step.

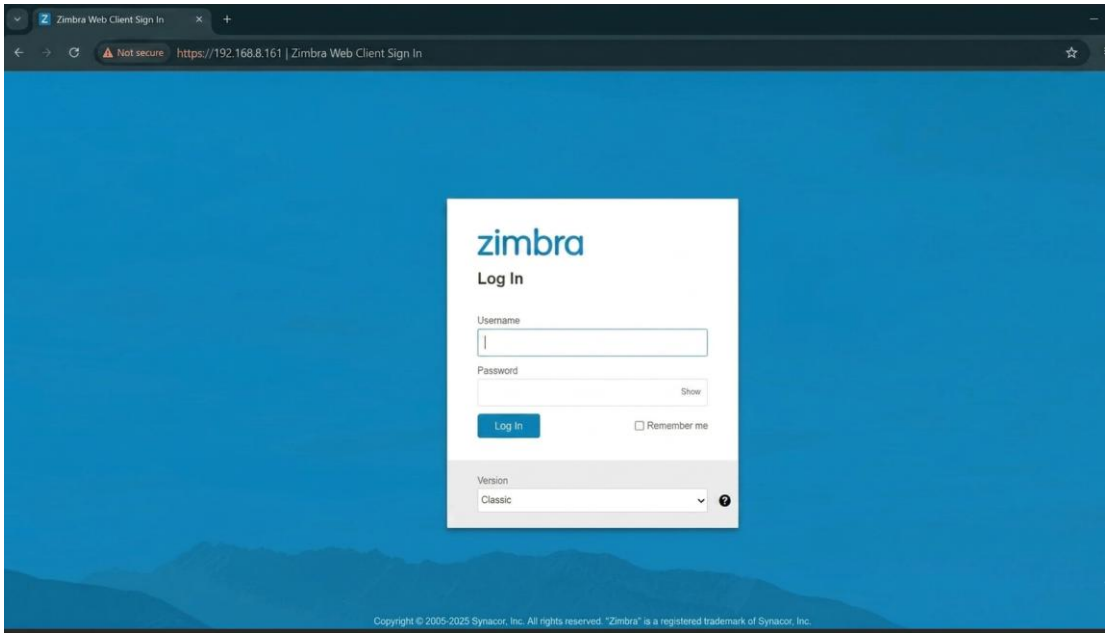


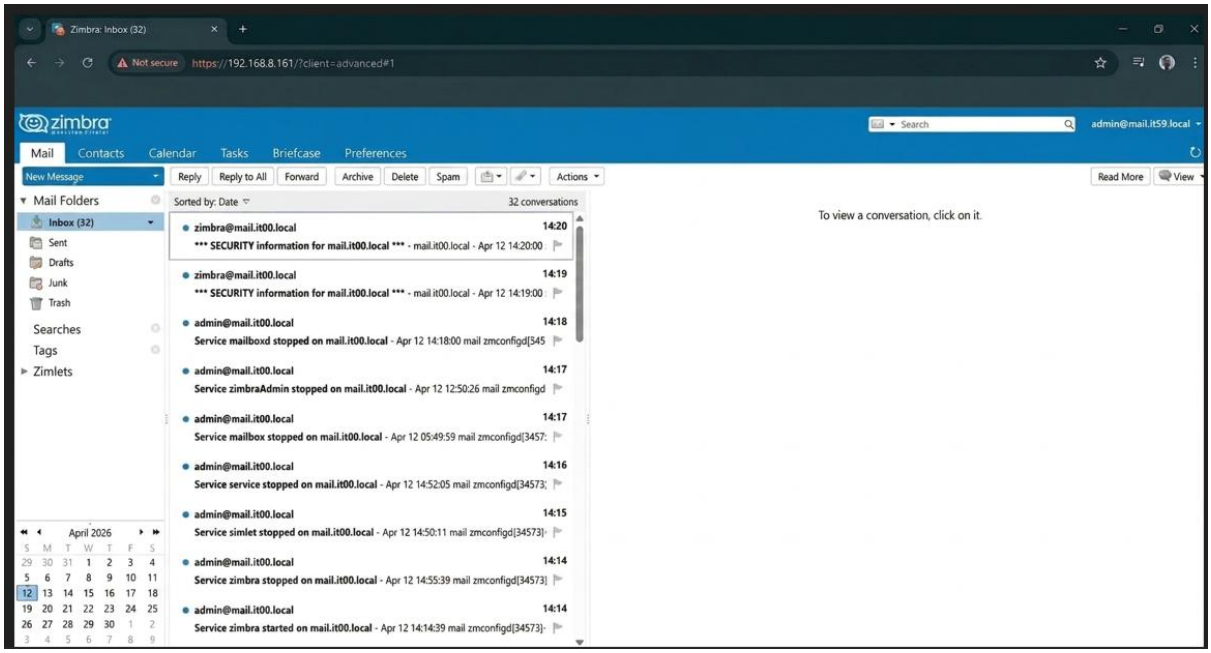
Above figure shows the list of mail accounts as well as the auto-generated service accounts from the Zimbra server using the it00.local domain. The screen capture proves that the user accounts kavindu@it00.local, kasun@it00.local, and sasindu@it00.local have been successfully set up and registered on the server. Besides the user accounts, some services like galsync and virus-quarantine have been generated by the Zimbra server, and they play an important role in the mail systems.

```
ubuntu@ubuntu:~$ ping mail.it00.local
PING mail.it00.local (192.168.0.1) 56(84) bytes of data.
64 bytes from mail.it00.local (192.168.0.1): icmp_seq=1 ttl=255 time=5.97 ms
64 bytes from mail.it00.local (192.168.0.1): icmp_seq=2 ttl=255 time=0.395 ms
64 bytes from mail.it00.local (192.168.0.1): icmp_seq=3 ttl=255 time=0.520 ms
64 bytes from mail.it00.local (192.168.0.1): icmp_seq=4 ttl=255 time=37.6 ms
64 bytes from mail.it00.local (192.168.0.1): icmp_seq=5 ttl=255 time=0.692 ms
^Z
[1]+  Stopped                 ping mail.it00.local
```

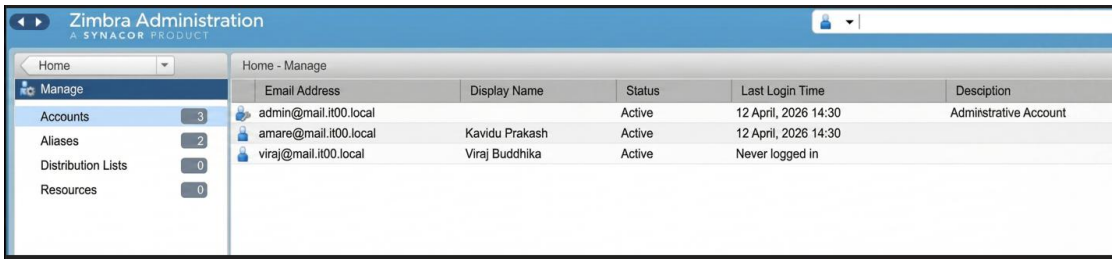
Above shows a successful ping test from the Ubuntu client to the mail server mail.it00.local. The hostname was correctly resolved to 192.168.0.1, which confirms that name resolution was working properly in the internal laboratory network. Multiple ICMP reply messages were received from the server, showing that the client could communicate with the mail server successfully over the host-only network.

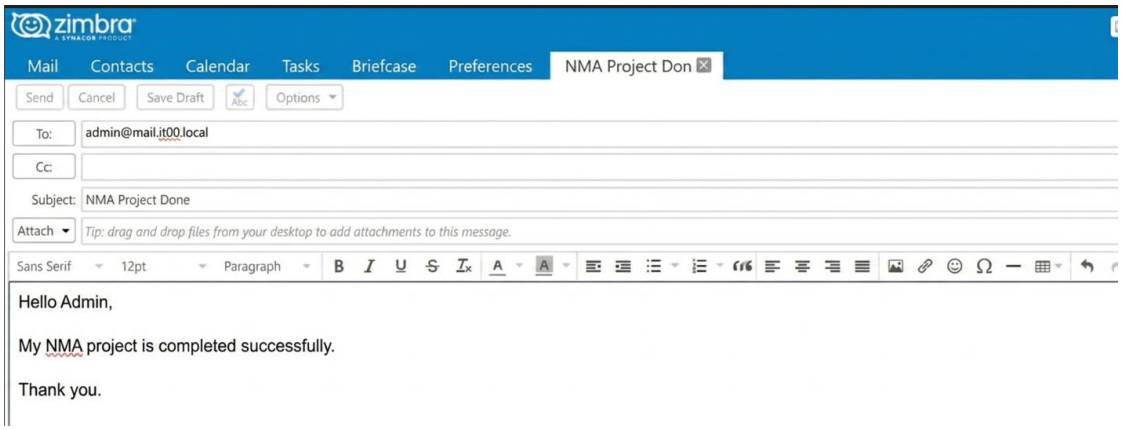
Zimbra mail proof





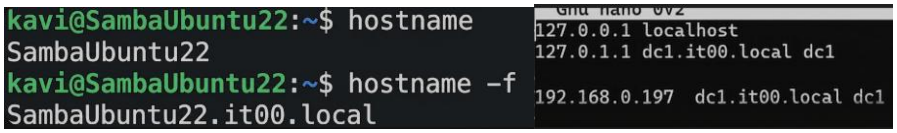
First figure illustrates the successful access to the Zimbra Web Client login interface via the server IP address. The second one demonstrates a successful login to the Zimbra mail system under the mail.it00.local domain, displaying the user inbox and confirming that the webmail service is properly configured and operational. This verifies that users are able to authenticate and access their email accounts through the Zimbra web interface.



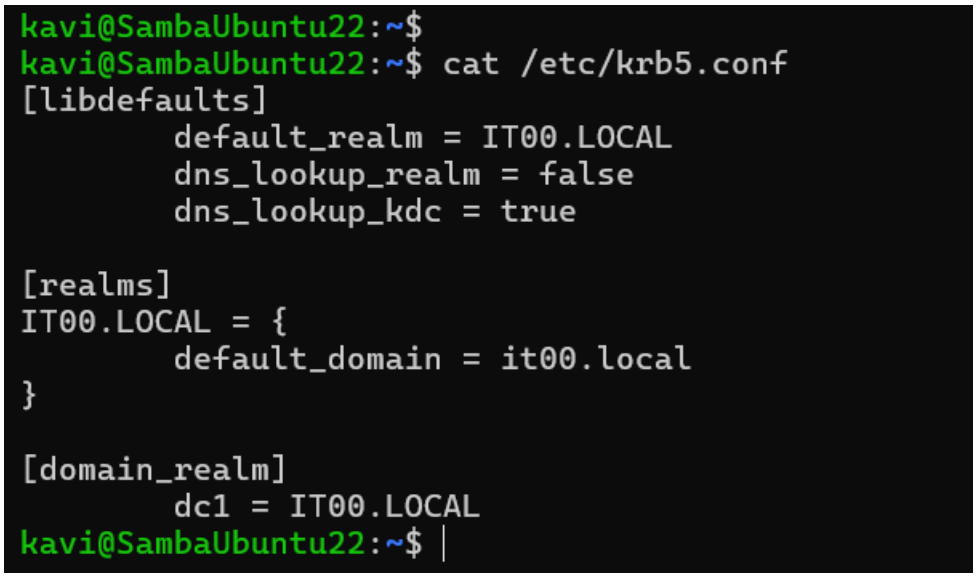


Step D – samba mail server

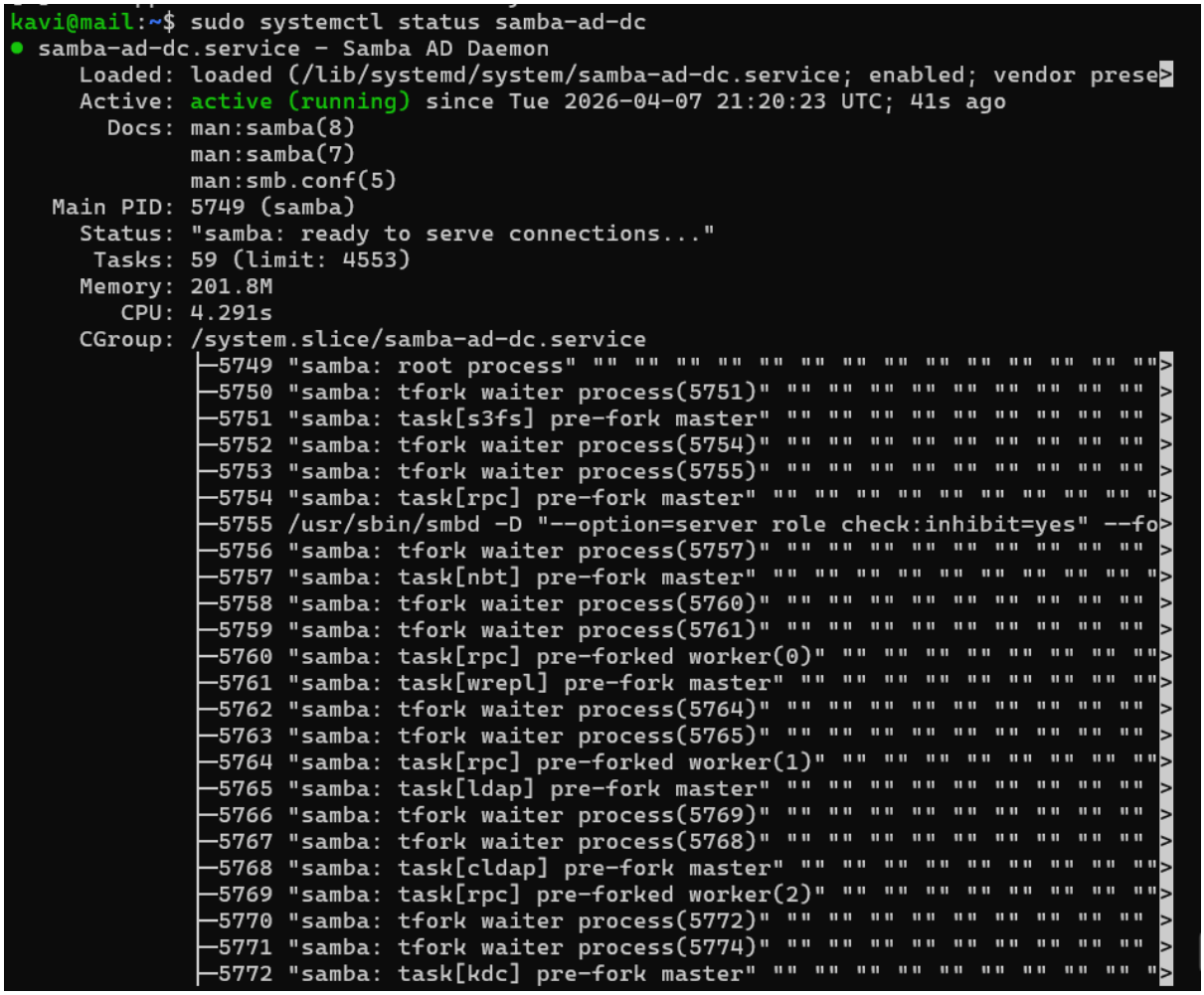
Samba is an open source software package that enables Linux servers to act like a Windows network by offering file and printer sharing along with directory services. For this task, the Samba software package has been used to configure an Active Directory Domain Controller for managing users, authenticating them and controlling access to their domains from clients in the it00.local network.



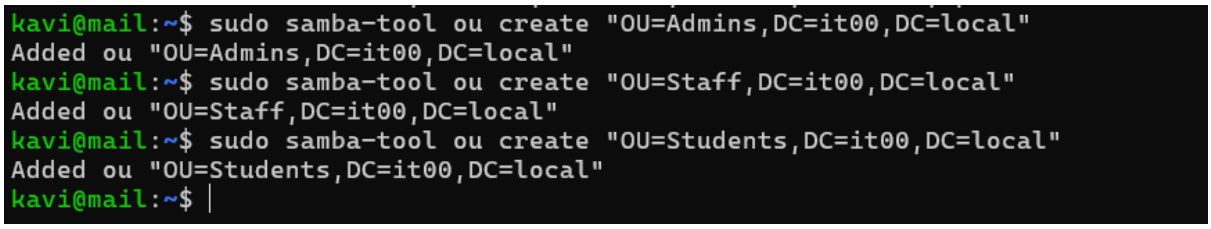
This figure confirms that the Samba server was configured with the hostname **dc1** and the fully qualified domain name **dc1.it00.local**. It also shows that the server identity was correctly associated with the internal IP address **192.168.0.197** through the local host mapping configuration. This step was essential to ensure reliable name resolution within the **it00.local** domain, which is required for the proper operation of directory, authentication, and domain controller services.



Kerberos configuration for the **IT00.LOCAL** realm in the Samba Active Directory environment.



It is clear from this picture that the Samba Active Directory Domain Controller was correctly launched, and the service was in an active running mode on the Ubuntu machine. From this information, it is clear that the Ubuntu machine was fully prepared to process all requests relating to domain functionality and provide directory services, authentication and administration in the it00.local domain. It was vital to do so to ensure that the domain controller functioned well and could support user authentication and administration in the lab setting.



```
kavi@mail:~$ sudo samba-tool user list
lasitha
viraj
ganindu
sasmitha
jayathu
ravi
Guest
krbtgt
sasindu
Administrator
kavindu
daniru
kavi@mail:~$
```

Above picture depicts the user account list from the Samba Active Directory Domain after completing the server's configuration process. It proves that more than one user account was successfully made under the domain such as lasitha, viraj, ganindu, sasmitha, jayathu, ravi, sasindu, kavindu, daniru. There were also some built-in user accounts which were needed for the directory service system. The existence of this user accounts indicates that the domain controller is working efficiently.

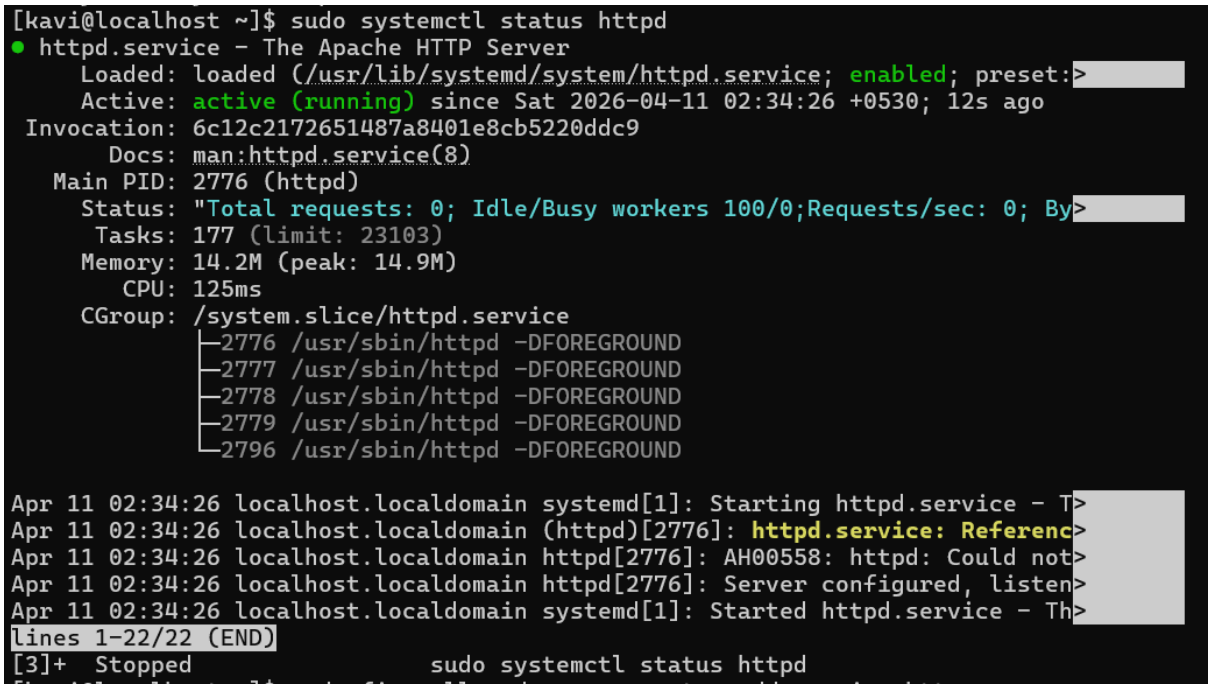
Part E - Web Server

DNS Active

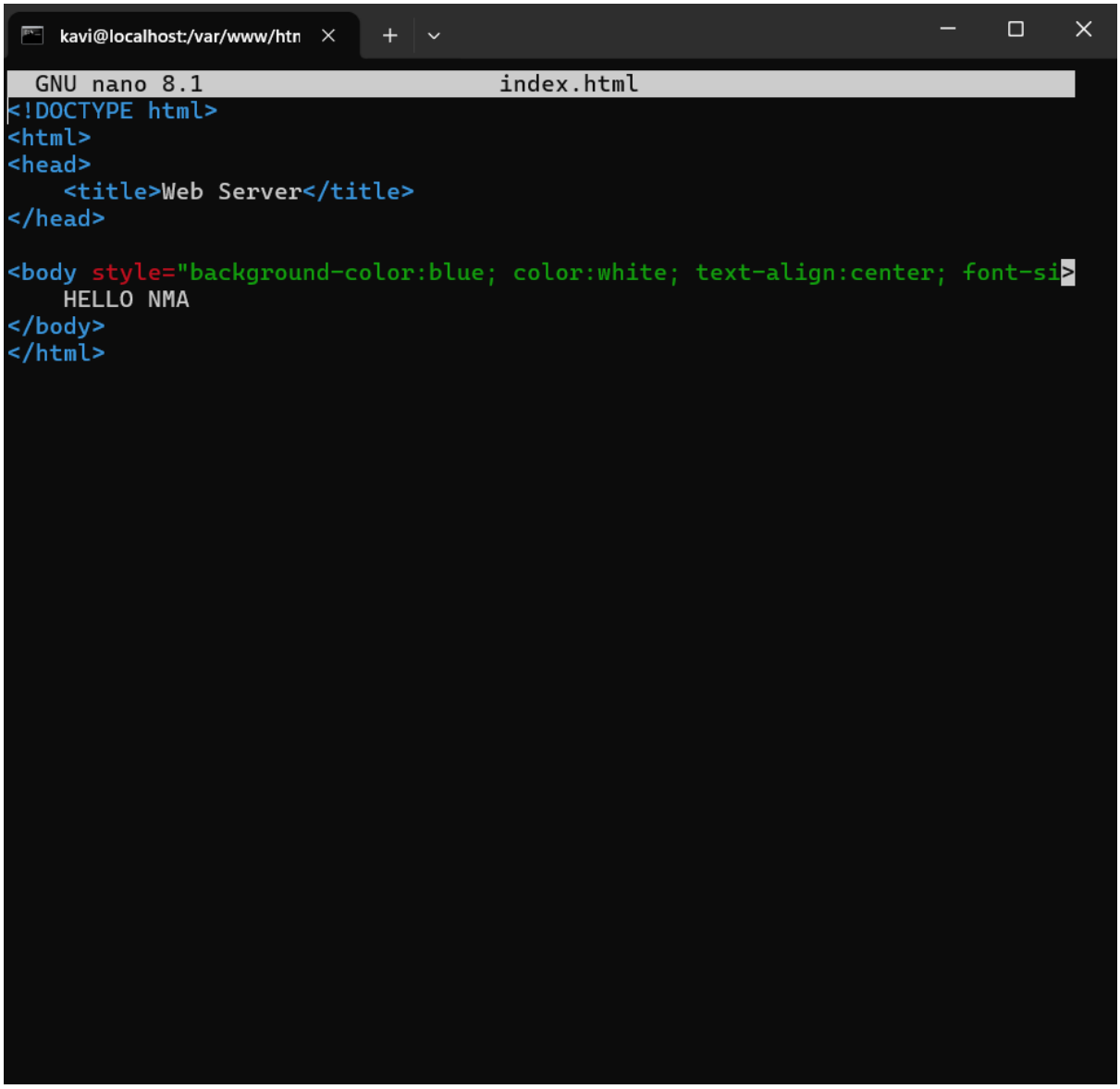
```
[root@localhost ~]# systemctl restart named
[root@localhost ~]# systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset>
     Active: active (running) since Sun 2026-04-12 22:03:02 +0530; 9s ago
 Invocation: 5ef311354e4d4dd19374652a984b9125
    Process: 2225 ExecStartPre=/bin/bash -c if [ ! "$DISABLE_ZONE_CHECKING">
    Process: 2229 ExecStart=/usr/sbin/named -u named -c ${NAMEDCONF} $OPTIO>
   Main PID: 2230 (named)
      Tasks: 10 (limit: 23103)
     Memory: 6.3M (peak: 7.3M)
        CPU: 81ms
     CGroup: /system.slice/named.service
             └─2230 /usr/sbin/named -u named -c /etc/named.conf

Apr 12 22:03:02 localhost.localdomain named[2230]: network unreachable reso>
Apr 12 22:03:02 localhost.localdomain named[2230]: network unreachable reso>
Apr 12 22:03:02 localhost.localdomain named[2230]: network unreachable reso>
Apr 12 22:03:02 localhost.localdomain named[2230]: network unreachable reso>
Apr 12 22:03:02 localhost.localdomain named[2230]: network unreachable reso>
Apr 12 22:03:02 localhost.localdomain named[2230]: network unreachable reso>
Apr 12 22:03:02 localhost.localdomain named[2230]: network unreachable reso>
Apr 12 22:03:02 localhost.localdomain named[2230]: network unreachable reso>
Apr 12 22:03:02 localhost.localdomain named[2230]: managed-keys-zone: Key 2>
Apr 12 22:03:02 localhost.localdomain named[2230]: managed-keys-zone: Key 3>
lines 1-23/23 (END)
```

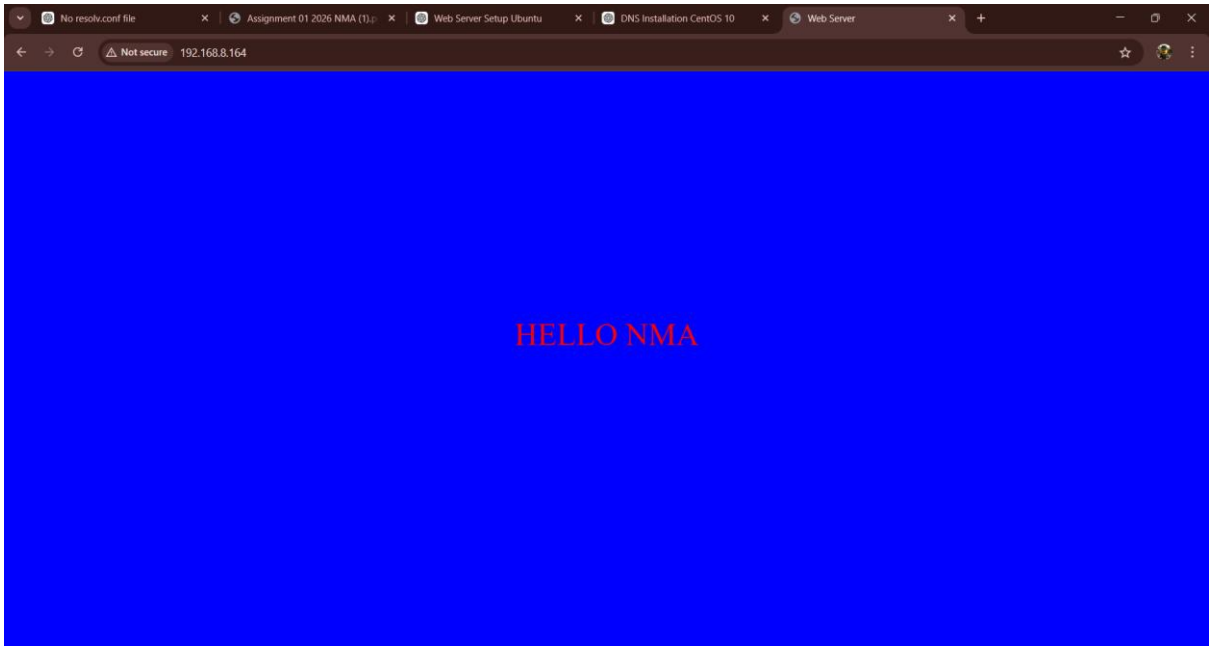
The above figure validates that the DNS server has been successfully started from the CentOS server and is running in an active mode. The above output reveals that the DNS daemon was successfully started, and the forward and reverse lookup zones have been properly loaded and put into action. This process of verification was essential since the DNS server plays a critical role in resolving the hostname in the laboratory setting, enabling computers to find each other through domain names rather than IP addresses alone.



This figure confirms that the Apache service was installed and running on the Ubuntu server. It verifies that the server platform required for web publication was active and available to serve HTTP requests. This is important because the web page requirement cannot be met unless the web server process is correctly installed and operational.



This figure shows the source content used to generate the required assignment web page. It verifies that the page was configured to display the exact text HELLO NMA and that the styling included the required blue background. This is important because the assignment evaluates both accessibility and correct page presentation, not merely the existence of a default web server page.



This figure provides the client-side validation of the published web service. It shows that the required HELLO NMA page was reachable from the client browser and displayed with the expected visual format. This is important because it confirms that the service was not only hosted locally on the server, but was accessible across the laboratory network as required by the assignment.

Summary

The completion of this task successfully indicated the creation and confirmation of a network services environment on the basis of Linux OS in the virtual lab. As part of the configuration of the environment, the use of the 192.168.0.0/24 network and the allocation of services among several virtual machines was applied for better reliability and avoidance of potential conflicts. For the Zimbra mail services, the Samba Active Directory service and Ubuntu servers were selected, whereas DNS and Apache Web Hosting were placed on the CentOS server. The correctness of the performed configuration was confirmed by successful verification through the appropriate network adaptation, internal host-only connection, and external administration by means of SSH connection.

The gathered evidence showed that all essential services functioned properly. Zimbra was installed and configured; user mail accounts were created and their connectivity was tested on the mail server side. Samba Active Directory was configured according to proper domain specifications, Kerberos realm and user accounts. In turn, DNS service worked correctly and properly loaded its zones, while Apache Web Hosting served web pages to client computers successfully.